

AUTOMATION RESEARCH REPORT

ALLIES CONNECT - GROUP 4

Abstract - Allies Connect is a mobile-first web platform developed by a team of students from Kennesaw State University for Allies Way, Inc. The application is designed to provide underprivileged people, volunteers, and nonprofit organizations within the state of Georgia a centralized platform to aid and better the community. The first phase of development created a manual minimal viable product, to establish a core foundation so future additions may be added without any drastic changes being needed. This report focuses on Phase II Automation Research and Roadmap Proposal. This report evaluates the feasibility, cost, risk, implementation effort, and user benefit of various potential automation features for future implementation into Allies Connect. These features focused on data accuracy and ingestion and minimizing administrative work and effort. The features were then scored based on the previously mentioned criteria and placed into priority tiers. The top four features were selected for the proposed Phase II Automation Roadmap. Pilot partner criteria was developed, along with success bench marks and risk mitigation strategies for initial testing and implementation of these features.

Background

Many underprivileged residents of Georgia are dependent on non-profit organizations, service providers, and other community outreach programs to access food, shelter, and additional necessary resources. Additionally, these organizations/providers are dependent on volunteers to operate their events, food drives, pop-ups, etc. Even though these organizations play an instrumental role in the community, it can be difficult to access information related to the events, volunteer opportunities, or the organizations themselves. Many organizations depend on social media posts, fliers, and word of mouth to spread awareness and gain support, which is unreliable and makes it difficult for people to keep track of various organizations' events and resources. Nonprofits also face many issues when it comes to manually maintaining general information, event and resources status, and volunteer shifts as it is tedious and repetitive.

Allies Connect is a mobile-first web platform that is designed to address these issues. This platform is developed for Allies Way, Inc. by a team of students at Kennesaw State University. It provides Georgia residents with a one-stop-shop solution for finding available and accurate resources, events, and volunteer opportunities from nonprofits and providers all across the state. Due to the application's ease-of-use, it is extraordinarily easy for these providers to actively publish, update, and coordinate events.

Purpose

The goal of the first phase for the development of Allies Connect was to produce a manual-first functional minimal viable product. This means that there are a minimal amount of automated features currently within the system and most actions are manually completed by a user. Manual tasks, especially ones that are repetitive, can take away valuable time from providers, administrators, and other users. While this is a large drawback for this first iteration, it has created a fundamental foundation for future development, including the incorporation of automated features. This is the goal of the second phase of Allies Connect: to reduce administrative burden and repetitive tasks for administrators and providers and improve data quality to enhance the user's experience by implementing automated features and workflows.

The purpose of this document is to provide an automation feasibility report for the second phase of Allies Connect. This automation feasibility report presents research related to how nonprofits currently publish data, how existing platforms use automated features, and what automated features can be incorporated into the system during the Allies Connect Phase II development. A prioritized set of automated feature recommendations along with implementation plans will also be provided in the Phase II Automation Roadmap.

Scope

The Automation Research Report evaluates six features identified in the [Automation Feasibility Research Plan](#). Calendar and event data ingestion is a feature that will allow the system to automatically import events and details from outside resources such as calendars and other data sources. Data classification is a feature that will autonomously and accurately categorize events, resources, and listings to enhance the searching capabilities and user experience. Automated volunteer reminders and scheduling allows for efficient communication with volunteers before and after their registered shifts. Resource availability verification allows for resources to be automatically checked to see if they are still active and up-to-date. Resource and event listing updates without manual intervention will drastically reduce the effort required to keep organizations information current. Lastly, stale data detection will automatically identify old or outdated data and inform administrators for review to prevent end users from being impacted. These features will be discussed and analyzed for future deployment in the Phase II of Allies Connect. Features beyond the ones mentioned above will need to be further assessed for future phases beyond Phase II.

Constraints

These recommendations were selected based on the constraints and end goals presented by Allies Way, Inc. in the Allies Connect project proposal and through discussions with the sponsors throughout the development of the Allies Connect Phase I MVP solution. The platform must use a free or Kennesaw State University-based infrastructure to minimize costs

for the end user. Any features implemented during the Phase I development must be adaptable to future developments of the application. Administrators must retain the ability to review and override any automated actions within the application, to ensure accuracy and trust throughout the website for all users. It is crucial that these constraints are followed so that this application can serve its true purpose in aiding those in need within the state of Georgia.

How Nonprofits Currently Publish Data

Nonprofits within the state of Georgia currently use a wide range of tools to share information regarding their resources, events, and volunteer opportunities. These tools consist of, but are not limited to, websites, social media accounts, physical mail pamphlets and flyers, static electronic documents, and phone number and email lists. All of these methods are fragmented and are difficult to implement together efficiently and effectively. These methods are also almost entirely manual, which can be tedious, difficult to maintain, and utilize due to the amount of time that is required to constantly post and update events as time goes on to various websites to spread awareness. This creates a large burden on the organization's administrator and restricts their time that can be utilized for more important things within the organization. Manually creating and managing events also introduces inconsistencies in data, misinformation, and general errors resulting in missed opportunities for aiding the community. Luckily, within these manual methods, patterns can be identified which allow for the implementation of a structured and consistent process. These types of processes are the solid foundation for implementing automated features.

Comparative Analysis of Existing Platforms

To fully understand the potential capabilities of implementing some automated features within Allies Connect, platforms within the realms of event management, personnel coordination, and data management were analyzed. This analysis focused on automated workflows, data ingestion, and the balance between automation and user intervention.

The first platform is Meetup. Meetup is a social event planning platform used for organizing group gatherings in a specified area. This company utilizes data-driven algorithms to recommend events that the user may like, depending on their selected preferences or past events they've signed up to attend. When it comes to creating events, Meetup supports the implementation of recurring event scheduling and autonomous event creation using iCalendar standard (ICS) and APIs.

Eventbrite is an event creation/promotion website that also manages the selling of tickets for said events. Eventbrite automates email and text message reminders, waitlists for events that have an attendance restriction, and analytics to better understand data and usage. Eventbrite also allows easy integration with satellite platforms using ICS and APIs.

Benevity is a platform used by large corporations to manage employee volunteer and charity programs. This platform helps companies internally track their employee involvement around the community and uses this data to better match employees with causes they feel more motivated to be involved with. Many of their features are automated but leave room for human integration with approval nodes for administrators to review.

HandsOn Connect is a volunteer management that tracks hours and impact to help nonprofit organizations with decision making to further their mission. The platform utilizes automated work flows like email confirmation after an account is created and reminders for events, but also relies on manual intervention for coordination and oversight.

211.org United Way is a platform that contains a vast directory of information regarding local resources and services. The system utilizes a standardized structure called Alliance of Information and Referral Systems (AIRS) which ensures consistent categorization for event and service data. With this standardization, they are able to use algorithm-based autonomous categorization for services. Additionally, the platform incorporates service validation and duplication detection, along with periodic review workflows to continuously validate information.

Candidate Automation Features

Calendar and Event Data Ingestion

Adding an autonomous calendar and event ingestion would allow users to upload public calendar URLs such as Google Calendar or some other ICS feed. Allies Connect would periodically check this URL for updates and automatically update or create event and resource listings. Incorporating this feature would decrease the amount of work required by providers as they would not have to insert the same event on multiple platforms. To ensure accuracy and trust, a provider approval node would be required for autopopulated events and resources. Many open-source libraries exist that would make incorporating this feature an easy task, including ical.js and ics.js. These libraries would be used within the backend of the application to enhance modularity and scalability. Events created by this feature will be placed in a provider approval similar to the pending organization approval for administrators, but providers will edit any data that needs to be changed.

Data Classification

When events or resources are created by providers, the feature will automatically suggest category types and classifications based on content within the title, description, provider name, or other component within the posting. This will increase the accuracy and consistency of the categorization of events and resources, while also saving time for the provider. As stated previously, these categories and classifications will be suggested and will require final approval by the provider before being added. The simplest version of this feature would involve key word matching utilizing Allies Connect category library. A more complicated,

yet still free, implication would involve a free-tier LLM API such as Google AI Studio, Groq, and OpenRouter. The restriction with this is the free-tier LLM APIs come with daily token limits. This would be an issue depending on what the expected daily usage would be for event and resource creation.

Volunteer Reminders and Scheduling

Volunteers that are registered for event shifts within the Allies Connect system currently receive automated shift sign up confirmations and reminders via email using Nodemailer and a custom timeout loop method. When volunteers register for an event shift, they will automatically be sent a confirmation email with the event name, shift time, and location. The volunteer will then be sent a reminder email at 8:00am the day before the event start time. The remaining component that needs to be implemented is a follow-up email asking for any feedback for the event. This feature can be implemented using Nodemailer, which is already being used within the application. This feature will be able to directly integrate with existing volunteer registration database tables but will require an additional workflow for the volunteer feedback.

Resource Availability Verification

This feature will consist of a regularly scheduled email to confirm the validity of a provider's resource. The email will show all of the resources details and prompt the provider to confirm that the data is still accurate and the resource is still active. Depending on the answer, the provider will be prompted to update the resource information. If the provider does not respond to the email within a certain time frame, the resource will be marked as inactive and be unavailable to site visitors. This feature can be implemented with Nodemailer, which is already being used within the application. Additional workflows will need to be created to deactivate resources.

Resource and Event Updates

The goal behind this feature is to create a fully autonomous system that discovers and updates resource and event listings through the scanning of public websites, social media posts, and other unstructured communications using free-tier LLM APIs. Ideally, this would reduce the time needed for providers to update information relating to resource and event listings. Implementing this feature would require a very well-trained model and potentially third-party assistance to accurately scrape public online data for information. To maximize the benefits and minimize the risks, a lot of time and money will be needed.

Stale Data Detection

This feature will function in a similar manner to the resource availability verification feature, but instead will identify listings and resources that have not been changed or modified within a specified time window. The system will perform this with a regularly scheduled query

into the resource and event databases and view the element specifying the last updated date and time. When a listing is marked as potentially stale, an email is sent to the associated provider to confirm the resource or event is still accurate. If the provider does not respond to inquiry within a specified window, the resource or event is marked as inactive, and an administrator is notified.

Feature Evaluation Matrix

Overall Scoring

An explanation of the evaluation matrix and scoring criteria can be found in the Automation Feasibility Research Plan document. Below shows a cumulative table of the evaluation results of the features.

Feature	Feasibility	Cost	User Benefit	Risk	Implementation Effort	Total Score	Priority Tier
Calendar and Event Data Ingestion	4	4	5	3	3	19	Should Have
Data Classification	3	3	4	3	3	16	Nice to Have
Volunteer Reminders and Scheduling	4	5	5	4	4	22	Must Have
Resource Availability Verification	4	4	5	4	3	20	Must Have
Resource and Event Updates	2	3	4	2	2	13	Not Recommended
Stale Data Detection	4	4	5	4	3	20	Must Have

Calendar and Event Data Ingestion

A four was given for the feasibility score because ical.js and ics.js can easily integrate with Node.js and the existing database design. Cost was given a score of four due to the only

incorporated costs being for development, testing, and deployment time. User benefit was given a score of five because it removes the effort from providers from needing to manually input events and resources that already exist on other calendars. Risk was given a score of three due to potential inconsistencies in ICS data, which can cause inaccuracies in event and information postings. This can be mitigated with the implementation of a review and approval node for providers or administrators. Finally, implementation effort was given a 3 due to the required experience and knowledge to identify ICS data feed edge cases.

Data Classification

Feasibility was given a score of three because the implementation of the AI-assisted approach would involve incorporating an external LLM API. The keyword approach would be easier to implement with the current system. Cost was given a score of three due to the limited access of the free-tier LLM APIs. If the API usage was known and below the free-tier daily limit, the cost score could increase. User benefit was given a score of four because accurately categorizing information for users would improve the searchability of events and listings for the end user. Risk was given a score of three due to the possibility of miscategorizing data, which would make finding an event or resource very difficult. This could be mitigated by adding a provider or administrator review node. Implementation effort was given a score of three due to the complexities of adding the necessary architecture and required testing to validate the accuracy of the LLM.

Volunteer Reminders and Scheduling

Feasibility was given a score four due to the already accessible volunteer data and incorporated Nodemailer library. Cost has a score of five due to the free libraries that will be used and easier implementation effort. User benefit was given a score of five due to the reduction in manual effort by providers to reach out to volunteers. Additionally, autonomously reminding volunteers can help increase attendance. Risk was given a score of four due to automated email incorporation is a well known feature in the software realm, so it will be easy to maximize the accuracy and quality. Implementation effort was given a score of four due to the vast documentation and support available for email and scheduling with Nodemailer.

Resource Availability Verification

Feasibility was given a score of four due to the accessible resource data and incorporated Nodemailer library. Cost was given a score of four due to the free Nodemailer library for sending emails. The reason the cost was not a five was just due to the increased development costs to adding additional workflows for updating or deactivating resources. User benefit was given a five due to making sure that resources are constantly available and the data that is being presented is accurate. Risk was given a score of four due to automated email incorporation is a well known feature. Implementation effort was given a three due to the need to add new workflows for updating the resources.

Resource and Event Updates

Feasibility was given a score of two due to autonomous web browsing and text parsing is not a current feature that is available given the system's current architecture. Cost was given a score of three due to the need for a high amount of LLM API token usage. Free-tier API daily limits will easily be exceeded if web scraping is constantly occurring to ensure data accuracy. User benefit was given a four due to the theoretical final product that would eliminate providers having to update events and resources, but this final product would be nearly impossible to reach. Risk was given a score of two due to the high risk of inaccurate data from unstructured sources and potential legal issues with unauthorized access to certain web pages. Implementation effort was given a score of two due to the high complexity, uncertain retain, and potential accuracy risks of this feature.

Stale Data Detection

Feasibility was given a score of four due to the accessible resource and event database tables and incorporated Nodemailer library. Cost was given a score of four due to the free Nodemailer library for sending emails. The reason the cost was not a five was due to the increased development costs to adding additional workflows for updating or deactivating events and resources. User benefit was given a five due to increased accuracy of resources and events provided by organizations. Risk was given a score of four due to automated email incorporation is a known feature and already present within the system. Implementation effort was given a three due to the need to add new workflows for resource and event updating and administrator nodes.

Recommendations and Prioritization

Must Have

The features which scored twenty or higher were listed as Must Have. This was due to how easily these features can be integrated with the current architecture and how highly the user would benefit from the implementation. It is highly recommended that the Volunteer Reminder and Scheduling, Stale Data Detection, and Resource Availability features be implemented first within the Phase II development of the Allies Connect project. As previously stated, all of these features address issues relating to repetitive manual tasks with potential data accuracy issues. Additionally, the main components for these features include Nodemailer and scheduled emailing. Being able to complete each of these features together would have the most impact on the project's development.

Should Have

The feature, Calendar and Event Data Ingestion, which scored between seventeen and nineteen was listed as Should Have. This feature also drastically helps reduce manual tasks by providers and administrators, but has a drawback with potential data accuracy issues due to potential inconsistent ICS data and edge cases not caught during testing. In an ideal world, this feature would also be listed as Must Have, but having four features required to be developed in the Phase II can come with time and capacity restrictions.

Nice to Have

The features with a score between fifteen and sixteen are listed as Nice to Have. Data Classification presents clear value to future development of the project and is technically achievable, but has at least one drawback due to implementation complexities. Data Classification was given a score of sixteen mostly due to its complexities with autonomous categorization of events and listings. Incorporating a LLM API can force architectural changes within the software and demand large amounts of testing to ensure consistency and accuracy. That being said, it would be best to implement this feature in a later development phase beyond Phase II, unless there is an unexpected availability in resources.

Not Recommended

The features with a score of fourteen and lower are listed as Not Recommended. The feature Resource and Event Updates was found to have high risk, high cost, and implementation complexity and effort. While the benefit of a fully completed and 100% accurate autonomous event and resource updating system would be immense, it is nearly impossible to get to that point considering the cost, time, and development capability restrictions. It is recommended that this feature not be pursued in future development phases.

Phase II Automation Roadmap

The [Phase II Automation Roadmap](#) can be found in the Allies Connect repository.

Risk Mitigation

There are many risks that have been identified through the feature evaluation process. Each risk is placed in one of three tiers of how likely the risk is to occur. These tiers are low, medium, and high.

Risk	Tier	Description	Mitigation
Data Decay	High	Automated features that create events and resources based on ingested data	Require provider or administrator approval nodes for autogenerated events and resources Implement override capabilities to disable these automated features
Exceeding Cost Limit	Low	Utilizing paid LLM APIs due to exceeding free-tier daily limits	Restrict to free alternatives such as keyword libraries
Provider Loss of Trust	Medium	Providers feel automated features are making incorrect changes or not properly importing information	Allow providers to opt-out of automated features within their provider dashboard
Third-Party Service Dependency	Medium	Leaning on external calendar structuring creates fragile and potential failures	Data ingestion design incorporates fail safes such as manual override or review to prevent faults from shutting system down
User Data Privacy Breach	Medium	Automated emails expose personally identifiable information to unauthorized users or entities	Minimize the personally identifiable information that is stored Use secure encryption and require user authentication

Success Metrics

When the automated features are implemented within the Phase II development, it is important to have criteria put in place to know whether the implementations are successful. Measurements need to be taken and data needs to be collected from the start of the Phase II development, throughout the pilot testing and production deployment periods.

Data Freshness

- At least eighty percent of active resources and events have been updated within the past sixty days

Administrative Workload

- Reduce number of manually written emails to volunteers from providers by at least eighty percent
- Reduce the amount of time spent by providers to manually create and update resources by at least eighty percent
- Reduce the amount of time by administrators manually inspected provider events and resources for outdated information by at least ninety percent

Volunteer Engagement

- Decrease volunteer absence rate by at least twenty percent
- Increase volunteer feedback for events by at least twenty percent

Sponsor Satisfaction

- Sponsor approval of Phase II Roadmap suggestions
- No reports of accuracy incidents from sponsor
- At least fifty percent of active providers utilize automated features within two months of full implementation

Conclusion

This automation research report for the Phase II development of Allies Connect identified various potential automation features that could highly benefit the Allies Connect platform as it continues to grow to help those in need within the state of Georgia. Through the comparative analysis and evaluation of six potential automation features using key criteria defined by the sponsor, these features were broken into 4 categories, must have, should have, nice to have, and not recommended. The top four features within the must have and should have categories were deemed highly beneficial for the reduction of administrative tasks and increase in data

quality and overall application experience. Per the request of the sponsor, these features utilize open-source and free libraries so that Allies Way, Inc. is not burdened with unnecessary costs. The recommended path forward is to start Phase II development with the implementation of the features that fall into the must have category as they have the lowest risk, cost, and implementation effort, while also being closely related and having the highest benefit. The next stage within Phase II would tackle the feature within the should have category. Success criteria metrics have been defined to measure the impact and results of the implementation of these features. It is strongly recommended that pilot testing be performed with a select group of organizations fitting the criteria mentioned in this report to ensure consistent and reliable results. Finally, sponsor approval of the Phase II Automation Roadmap is required to proceed with future development to validate its alignment with Allies Way Inc. goals.

References

Corporate volunteer opportunities: Volunteering. Benevity. (n.d.).

<https://benevity.com/products/volunteer>

Creating a repeating event – Meetup. (n.d.).

<https://help.meetup.com/hc/en-us/articles/39795590048781-Creating-a-repeating-event>

Home. HandsOnConnect. (2023, January 25). <https://www.handsonconnect.com/>

Homepage. Call 211 for Essential Community Services | United Way 211. (n.d.).

<https://www.211.org/>

Inform USA standards. Standards - Inform USA (formerly AIRS, the Alliance of Information and Referral Systems). (n.d.). <https://www.informusa.org/standards>

What emails will attendees automatically receive?. Eventbrite Help Center. (n.d.).

<https://www.eventbrite.com/help/en-us/articles/748624/what-emails-will-attendees-automatically-receive/>